
RaoRocketSim

A Differentiable Rao Thrust-Optimized-Parabola Rocket-Nozzle Design Toolbox

Technical Reference — Methodology, Mathematics, and Physics

This reference compiles the methodology, governing equations, software architecture, validation status, and roadmap of the RaoRocketSim project. Every symbol, ratio, and governing equation has been cross-checked against the source code and the primary literature in `propulsion_texts/`.

Engineering status. Every generated contour is stamped `hardware_qualified = False`. Passing the repository's design gates means a result cleared internal preliminary screening only. It does **not** replace independent CFD, conjugate heat-transfer analysis, structural FEA, combustion-stability work, material allowables, manufacturing review, inspection, proof testing, or hot-fire qualification.

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Reference generated June 2026

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1 Overview

RaoRocketSim is a Python research and preliminary-design toolbox for axisymmetric rocket-nozzle aerodynamics. It pairs a practical Rao thrust-optimized parabolic (TOP) bell workflow with quasi-one-dimensional performance estimation, low-order engineering screens, an axisymmetric method-of-characteristics (MOC) solver, a finite-dimensional Rao variational boundary-value problem (BVP), a vendored NASA/JHU reference-code port, and a differentiable JAX backend with exact-Jacobian solves and gradient sensitivities.

The **default, trusted** path is the chart-based Rao/TOP quadratic-Bézier contour: it is deterministic, endpoint-exact, and benchmarked against published geometry. The MOC and variational solvers are **active research implementations** — they expose residuals, topology, reference comparisons, and explicit reliability metadata, but they are not promoted to design-validated or hardware-qualified status.

1.1 Capabilities and maturity

Area	Current implementation	Status
Rao/TOP geometry	Upstream + downstream throat arcs and a quadratic Bézier bell using interpolated Rao-chart angles	Trusted baseline
Ideal performance	Constant- γ , calorically perfect, quasi-1D isentropic flow; C_F , thrust, I_{sp} , c^* , $\dot{m}$, exit state	Preliminary
Thermochemistry	Built-in combustion-product constants, or optional RocketCEA chamber γ , M_w , T_c , c^*	Preliminary (constant- γ flow)
Direct MOC optimization	Axisymmetric characteristic march coupled to a monotone spline wall under SLSQP / Nelder–Mead	Experimental
Rao variational / MOC BVP	NASA-topology seed, Rao stationarity, characteristic compatibility, mass + length closure, D -state continuity, validity diagnostics	Experimental research path
Differentiable solver	JAX/Optimistix Levenberg–Marquardt residual solve, differentiable kernel march, optional solved θ_B , exact C_F sensitivities	Research-grade
Wall pressure & separation	Quasi-1D wall pressure; Summerfield, Kalt–Badal, Schmucker screens	Screening only
Thermal / cooling / structure	Boundary-layer displacement, Bartz-style heat flux, regen-channel, thin-wall hoop-stress screens	Screening only
Geometry export	CSV, revolved STL, STEP (CadQuery B-rep or faceted AP214), Inventor IPT manifest	CAD-review input
Validation	Unit/regression tests, literature manifests, NASA/JHU parsers, parity checks	Software verification

The maturity column is enforced in code: each contour carries a `design_status` keyed by method (`preliminary_top_geometry`, `experimental_moc_geometry`, `experimental_variational_geometry`, `experimental_rao_variational_moc_bvp`) and a research-grade `ContourReliability` level (§11.3).

1.2 Test and validation status

The most recently recorded run of the normal (non-slow) selection, on **June 14, 2026**, reports:

```
784 passed, 4 xfailed, 26 deselected in 350.73 s
```

produced with

```
MPLCONFIGDIR=/tmp/raosim-mpl \
.venv-jax/bin/python -m pytest -q -m "not slow"
```

The repository contains 339 test functions, expanded by parametrization to the counts above. The four expected `xfails` record known research gaps: the unresolved provenance of one historical NASA TT' fixture, an unpackaged Cuffel–Back–Massier dataset, and literature-promotion tests for the experimental MOC and legacy variational paths. The 26 deselected tests are long JAX solves, convergence studies, NASA fixed-end closure, and the full Rao-chart sweep. This is *software and mathematical verification*, not physical validation against experiment or CFD.

2 What the tool does

Given a chamber pressure, ambient pressure, throat size or target thrust, expansion ratio, propellant model, and contour method, RaoRocketSim can:

- generate a reduced-length Rao/TOP bell, a conical nozzle, an MOC-optimized bell, or an experimental Rao variational/MOC contour;
- size throat radius from a requested thrust, or size expansion ratio for matched ideal expansion;
- compute ideal exit Mach number and pressure, thrust coefficient, thrust, mass flow, effective exhaust velocity, and specific impulse;
- estimate wall pressure, overexpansion separation, altitude performance, boundary-layer displacement, heat flux, regenerative-cooling capacity, and thin-wall pressure stress;
- generate simplified chamber and convergent geometry from L^* and contraction ratio;
- sweep ε , P_c , or R_t ; compare contour families; and run literature-backed benchmark cases with explicit pass / report / xfail policies;
- plot contours, characteristic nets, Mach / pressure / angle fields, wall distributions, exit-plane profiles, topology, residual diagnostics, and JAX sensitivity fields;
- export versioned CSV, STL, STEP, JSON, Markdown, and metadata artifacts;
- compute exact derivatives of the control-surface C_F with respect to the solved node variables via the JAX backend.

3 Physical model

3.1 Assumptions

The core gas-dynamics and nozzle solvers assume steady, inviscid, adiabatic flow; a calorically perfect ideal gas with constant γ ; isentropic expansion except where an empirical loss or separation screen is applied; axisymmetry for the active MOC and Rao implementations; and a choked throat with a fully supersonic divergent section. There is **no** reacting-flow chemistry evolution, finite-rate kinetics, particles, film cooling, wall roughness, ablation, embedded shocks, side loads, or fluid–structure

interaction inside the solved flowfield. The optional CEA integration supplies a chamber-property *snapshot*; the nozzle flow is still evaluated with a single effective chamber γ .

3.2 Propellant / thermochemistry data

The built-in database stores nominal *combustion-product* properties (the exhaust gas at the nominal mixture ratio), not raw propellant properties:

Propellant	γ	M_w [kg/mol]	T_c [K]	η_{Isp}	O/F
N2O/Ethanol	1.22	0.0260	2800	0.92	5.5
LOX/RP-1	1.23	0.0235	3400	0.96	2.6
LOX/LCH4	1.24	0.0220	3500	0.96	3.5
LOX/LH2	1.20	0.0100	3250	0.98	6.0

The specific gas constant is $R = R_u/M_w$ with $R_u = 8314.46 \text{ J kmol}^{-1}\text{K}^{-1}$. Users may also supply custom properties or request RocketCEA-derived chamber values.

3.3 Isentropic gas dynamics

For Mach number M and ratio of specific heats γ , the implemented stagnation relations are

$$\frac{T}{T_0} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{-1}, \quad \frac{p}{p_0} = \left(\frac{T}{T_0}\right)^{\gamma/(\gamma-1)}, \quad \frac{\rho}{\rho_0} = \left(\frac{T}{T_0}\right)^{1/(\gamma-1)}. \quad (1)$$

The area–Mach relation is

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2}M^2\right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}, \quad (2)$$

inverted with Newton iteration on the chosen (subsonic or supersonic) branch. The Prandtl–Meyer function and Mach angle are

$$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}} \tan^{-1} \sqrt{\frac{\gamma-1}{\gamma+1} (M^2 - 1)} - \tan^{-1} \sqrt{M^2 - 1}, \quad \mu = \sin^{-1} \left(\frac{1}{M} \right), \quad (3)$$

with $\nu(M)$ inverted by Newton iteration using $d\nu/dM = \sqrt{M^2 - 1} / [M(1 + \frac{\gamma-1}{2}M^2)]$. These are standard compressible-flow results (Anderson, *Modern Compressible Flow*; cross-checked against `propulsion_texts/prmeyer.pdf`).

3.4 Performance model

With $\varepsilon = A_e/A_t$, $A_t = \pi R_t^2$, and $A_e = \varepsilon A_t$, the ideal one-dimensional thrust coefficient is

$$C_F = \sqrt{\frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{p_e}{p_c}\right)^{\frac{\gamma-1}{\gamma}}\right]} + \left(\frac{p_e - p_a}{p_c}\right) \varepsilon. \quad (4)$$

The built-in propellant model applies an empirical efficiency multiplier η_{Isp} to the complete ideal coefficient:

$$C_{F,\text{actual}} = \eta_{Isp} C_F, \quad F = C_{F,\text{actual}} p_c A_t. \quad (5)$$

The characteristic velocity, mass flow, specific impulse, and effective exhaust velocity are

$$c^* = \frac{\sqrt{\gamma R T_c}}{\gamma \sqrt{(2/(\gamma+1))^{(\gamma+1)/(\gamma-1)}}}, \quad \dot{m} = \frac{p_c A_t}{c^*}, \quad I_{sp} = \frac{C_{F,\text{actual}} c^*}{g_0}, \quad V_e = I_{sp} g_0, \quad (6)$$

with $g_0 = 9.80665 \text{ m s}^{-2}$.

These are **ideal-cycle** estimates. The efficiency multiplier is a single lumped factor, not a resolved loss model, and must not be read as a prediction of combustion, boundary-layer, two-phase, or chemical losses. (Formulation: Sutton & Biblarz, *Rocket Propulsion Elements*; Anderson.)

4 Contour methods

4.1 Conical reference

The conical utility reuses the bell throat arcs and a straight divergent wall at half-angle α . Its length and classical divergence factor are

$$L_{\text{cone}} = \frac{R_e - R_t}{\tan \alpha}, \quad \eta_{\text{div}} = \frac{1 + \cos \alpha}{2}. \quad (7)$$

The comparison module also estimates a bell divergence loss from an assumed linear exit-plane flow-angle profile,

$$\eta_{\text{div}} \approx \frac{\int_0^{R_e} \rho u_x |\mathbf{u}| r \, dr}{\int_0^{R_e} \rho |\mathbf{u}|^2 r \, dr}, \quad C_{F,2D} \approx \eta_{\text{div}} C_{F,1D}, \quad (8)$$

which is a comparison aid, not a resolved exit-plane solution of the Bézier contour.

4.2 Rao/TOP Bézier baseline (default, trusted)

The baseline contour has three pieces: an upstream circular throat arc ($R_u = 1.5 R_t$), a downstream circular throat arc ($R_d = 0.382 R_t$), and a quadratic Bézier bell from inflection point N to exit point E . The exit radius and reference 15° cone length are

$$R_e = R_t \sqrt{\varepsilon}, \quad L_{15} = \frac{R_e - R_t}{\tan 15^\circ}, \quad L_n = \frac{L_{\%}}{100} L_{15}, \quad (9)$$

and the bell is

$$\mathbf{B}(t) = (1-t)^2 \mathbf{N} + 2(1-t)t \mathbf{P}_1 + t^2 \mathbf{E}, \quad 0 \leq t \leq 1, \quad (10)$$

where $\mathbf{P}_1$ is the intersection of the tangent leaving N at angle θ_n and the tangent entering E at angle θ_e .

Chart angles and provenance. The default (θ_n, θ_e) are bilinearly interpolated from embedded Rao/NASA chart tables spanning approximately $4 \leq \varepsilon \leq 50$ and $60\% \leq L_{\%} \leq 100\%$. The tables reproduce Rao's TOP design charts (Rao, *Approximation of Optimum Thrust Nozzle Contour*, ARS J. 30(6), 1960; reproduced in Sutton and NASA SP-8120). Rao's underlying optimum study was computed for $\gamma = 1.23$; per Rao (*Recent Developments in Rocket Nozzle Configurations*, ARS J. 31(11), 1961), the optimal *contour* is nearly γ -insensitive at fixed (ε, L) — only C_F depends strongly on γ — so the angle tables remain valid comparison targets at other γ . This method is the trusted baseline because it is deterministic, smooth, endpoint-exact, benchmarked against explicit TOP geometry, and does not depend on the convergence of an experimental flow solver.

4.3 Chamber and convergent geometry

Optional upstream geometry is sized from contraction ratio $CR = A_c/A_t$ and characteristic length $L^* = V_c/A_t$:

$$R_c = R_t \sqrt{CR}, \quad V_c = L^* A_t. \quad (11)$$

The code subtracts the convergent conical-frustum volume and assigns the remainder to a cylindrical chamber. This is a geometric volume model only.

4.4 Direct MOC wall optimization

This path parameterizes the bell as a monotone cubic Hermite spline with decision variables $\mathbf{q} = [\theta_n, r_1, r_2, \dots, r_{nc}]$. For each candidate wall it builds a transonic starting line, marches an axisymmetric characteristic net with wall feedback, samples the exit plane, and minimizes a cost combining negative exit thrust, exit-flow-angle penalties, radius monotonicity, and curvature regularization. SciPy SLSQP is used when available, with a NumPy Nelder–Mead fallback. The exported bell is currently reconstructed as a smooth quadratic Bézier from the optimized entrance and exit angles — one reason it remains `experimental_moc_geometry`.

4.5 Rao variational / MOC BVP (main research solver)

This solver seeds a finite-dimensional BVP from the NASA/JHU topology

$$TT' \rightarrow B \rightarrow BD \rightarrow D \rightarrow DE \rightarrow E. \quad (12)$$

Its default configuration uses the JAX/Optimistix Levenberg–Marquardt backend with exact autodiff Jacobians, the characteristic residual formulation, a NASA-style fixed-end topology seed, full position / flow-angle / Mach continuity at point D , and a continuation ladder that raises weights on the mass, length, and endpoint constraints. The solved control-surface unknowns are, schematically,

$$\mathbf{u} = [M_i, \theta_i, r_i \mid \lambda_2, \lambda_3, \log C, f_D], \quad (13)$$

with optional wall unknowns and an optional live kernel angle θ_B . Here f_D is the arc-length fraction locating D on the kernel characteristic BD .

4.5.1 Axisymmetric characteristics

With nodes ordered downstream and the axisymmetric source $S = \sin \theta \sin \mu / r$, the implemented compatibility relations are

$$C^+ : \quad \frac{dr}{dx} = \tan(\theta + \mu), \quad d(\theta - \nu) = -S ds, \quad (14)$$

$$C^- : \quad \frac{dr}{dx} = \tan(\theta - \mu), \quad d(\theta + \nu) = +S ds. \quad (15)$$

The C^+ control surface DE is enforced geometrically by reconstructing its axial coordinates,

$$x_{i+1} = x_i + \frac{r_{i+1} - r_i}{\tan(\bar{\theta}_i + \bar{\mu}_i)}, \quad x_1 = x_D(f_D). \quad (16)$$

The source term and $C^\pm$ invariant pairing are taken from Anderson (*Modern Compressible Flow* §11.4) and Zucrow & Hoffman (*Gas Dynamics* Vol. 2, Ch. 17), and are oracle-validated against the NASA M3.5Perf grids (the corrected forms hold at RMS $\sim 10^{-6}$ – 10^{-8} on kernel and Deriv characteristics).

4.5.2 Rao stationarity

The critical Mach number is

$$M^* = \sqrt{\frac{(\gamma + 1)M^2}{2 + (\gamma - 1)M^2}}. \quad (17)$$

The primary algebraic optimum-thrust condition along the control surface DE is

$$M^* \frac{\cos(\theta - \alpha)}{\cos \alpha} = C, \quad \alpha = \mu = \sin^{-1}(1/M), \quad (18)$$

implemented in logarithmic form for conditioning. This is Eq. (1) of Rao, Beck & Booth, AIAA 99-2584 ([propulsion_texts/rao1999.pdf](#)). Its differential identity, Eq. (2) of the same paper,

$$d \ln M^* - (d\theta - d\alpha) \tan(\theta - \alpha) + d\alpha \tan \alpha = 0, \quad (19)$$

is retained as a secondary consistency check. The older direct-method functionals are also available; per unit radial extent their stagnation-normalized forms are

$$f_1 = 2\pi r \left[\left(\frac{p}{p_0} - \frac{p_a}{p_0} \right) + \frac{\rho}{\rho_0} \gamma M^2 \frac{T}{T_0} \frac{\sin(\phi - \theta) \cos \theta}{\sin \phi} \right], \quad (20)$$

$$f_2 = 2\pi r \frac{\rho}{\rho_0} \bar{V} \frac{\sin(\phi - \theta)}{\sin \phi}, \quad f_3 = \cot \phi, \quad (21)$$

representing axial thrust, mass flow, and axial length respectively.

4.5.3 Mass and endpoint closure

Mass closure compares the surface-normal flux integral on DE and the selected BD segment,

$$\dot{m}_\Gamma = \int_\Gamma 2\pi r \rho V |\sin(\beta - \theta)| ds, \quad \dot{m}_{DE} = \dot{m}_{BD}. \quad (22)$$

The BVP additionally pins the commanded exit station and radius, $x_E = L_n$ and $r_E = R_t \sqrt{\varepsilon}$, and reports scaled residuals for mass, length, stationarity, characteristic compatibility, geometry, regularization, penalties, wall endpoints, and wall tangency.

4.5.4 Smooth-flow validity

The classical Rao validity inequality is evaluated along DE :

$$b = 1 - \frac{d\alpha}{d\theta} \frac{\tan(\theta - \alpha) + \tan \alpha}{\tan(\theta - \alpha) - \tan \alpha} \geq 0. \quad (23)$$

If b drops below tolerance, the requested smooth variational solution lies outside the shock-free Rao region; the construction is flagged `invalid_short_nozzle_region` and the reliability level is downgraded. This boundary relation is the perfect-gas optimum-thrust boundary of Rao, Beck & Booth (1999) and is discussed in Östlund's KTH thesis ([propulsion_texts/fulltext01.pdf](#), §3).

For the repository's $\varepsilon = 10$, 80%-length, $\gamma = 1.4$ regression case, the smooth stationary- DE reference is

```
theta_B = 25.5659 deg
f_D = 0.15216
D = (M = 3.40145, theta = 18.5182 deg)
E = (M = 3.47655, theta = 11.1193 deg)
```

a numerical regression point for the implemented formulation, not a universal nozzle-design result.

4.6 Transonic starting lines

The MOC code provides several throat starting-line models: `kliegel_levine` (third-order toroidal-coordinate series; default), `sauer_modified` (compact leading-order curved-throat approximation), `area_ratio` (quasi-1D area–Mach line), `hall` (deprecated alias for `sauer_modified`), and `nasa_visible_kliegel_levine` (a source-faithful NASA/JHU compatibility mode used by the reference port). The theory-correct Kliegel–Levine path documents and tests several coefficient-transcription corrections; the source-faithful mode deliberately preserves the historical C++ integer-division behavior required for NASA binary/output parity.

5 Differentiable (JAX) backend

`raosim/jax/` provides the default inner solver for the Rao BVP and the gradient API. The design keeps a single boundary: the NumPy shell in `rao_variational.solve_rao_bvp` still owns seeding, kernel construction, reliability gating, diagnostics, and output assembly, while the JAX layer owns only the inner least-squares solve.

- **Solver.** Optimistix Levenberg–Marquardt with the *exact* `jacfwd` Jacobian of the assembled residual, replacing SciPy’s finite-difference `least_squares`. Box bounds are handled by a smooth reparametrization $u = \text{lo} + (\text{hi} - \text{lo}) \sigma(z)$ so every iterate stays strictly inside the box. A constraint-weight continuation ladder up-weights the single-element mass/length/endpoint residuals; the reported residual is always the unweighted one, so reliability gates stay comparable across backends.
- **Differentiable kernel march and θ_B .** The NASA-style kernel march and the live θ_B secant path are differentiable.
- **Sensitivities.** `rao_sensitivities(config, solution=...)` returns exact derivatives of the control-surface C_F with respect to the solved node variables, explicit fixed-solution partials with respect to p_a/p_0 and γ , and the residual-Jacobian condition number.

Not yet provided (v2 deferrals): total re-solved (implicit-function) design derivatives with respect to R_t , ε , length percentage, or γ ; a differentiable sensitivity map carried through to the final manufacturable bell wall; and the Hessian. The pinned, tested stack is `jax==0.6.2`, `jaxlib==0.6.2`, `optimistix==0.0.11`, `equinox==0.13.8`.

6 Screening models

These models are deliberately low-order: useful for ranking concepts and rejecting unsuitable designs, **not** for qualification.

6.1 Wall pressure

At each contour radius the code sets $A(x)/A_t = (r(x)/R_t)^2$, inverts the area–Mach relation, and evaluates $p(x)/p_c$ isentropically. A positive downstream pressure increment is flagged as non-monotonic (motivated by NASA SP-8120).

6.2 Separation

Three empirical overexpansion-separation criteria are exposed (as *implemented*):

$$\begin{aligned}
 p_{\text{sep}} &\approx 0.4 p_a && \text{(Summerfield),} \\
 \frac{p_{\text{sep}}}{p_a} &\approx \frac{1}{1.88 M_e - 1} && \text{(Kalt-Badal),} \\
 \frac{p_{\text{sep}}}{p_c} &\approx \left(\frac{p_a}{p_c}\right)^{0.8} M_e^{-1} && \text{(Schmucker).}
 \end{aligned} \tag{24}$$

The first contour station whose quasi-1D wall pressure falls below the chosen threshold is reported as the estimated separation location. Side loads, shock separation, hysteresis, and transient startup are not modeled. (Criteria surveyed in NASA SP-8120 and Stark, 2009.)

6.3 Boundary layer

The displacement-thickness screen uses a turbulent flat-plate-style correlation

$$\delta^* \approx 0.046 \frac{s}{Re_s^{1/5}} \sqrt{\frac{T}{T_w}}, \tag{25}$$

and estimates an effective exit area ratio from $r_e - \delta_e^*$. Gas viscosity uses a Sutherland-type estimate.

6.4 Heat flux and cooling

The heat-flux model preserves Bartz-like sensitivity to chamber pressure, characteristic velocity, throat diameter, gas temperature, and wall temperature, then applies empirical axial and area scaling (`bartz_style_screening` — not a full Bartz implementation). For regenerative cooling, total absorbed heat and coolant rise are estimated by

$$\dot{Q} = \int q''(x) 2\pi r(x) dx, \quad \Delta T_c = \frac{\dot{Q}}{\dot{m}_c c_{p,c}}, \tag{26}$$

followed by a one-resistance wall/convection temperature estimate. Channel pressure drop, boiling, coking, critical heat flux, rib conduction, and property variation are not solved.

6.5 Structural screen

The pressure-stress estimate is the thin-wall hoop relation

$$\sigma_h \approx \frac{p_c r_{\text{max}}}{t_w}. \tag{27}$$

Reported margins compare this stress, the estimated wall temperature, and the peak heat flux against user-supplied material limits. Temperature-dependent allowables, fatigue, creep, weld efficiency, buckling, and combined loads are outside the model.

6.6 Atmosphere and trajectory

Altitude performance uses piecewise ISA layers (troposphere -6.5 K/km from 0–11 km, isothermal 11–20 km, $+1.0$ K/km 20–47 km) with an exponential tail above 47 km. The optional trajectory module integrates a 1-D vertical point mass,

$$m \dot{v} = F - \frac{1}{2} \rho v |v| C_D A - m g(h), \quad g(h) = g_0 \left(\frac{R_E}{R_E + h} \right)^2, \quad (28)$$

with $R_E = 6.371 \times 10^6$ m. It is not wired into the main CLI and does not model guidance, pitch, staging, throttling, winds, or 6-DOF dynamics.

7 Installation

Python 3.12 is recommended because the differentiable backend is pinned to a tested JAX stack.

```
python3.12 -m venv .venv-jax
source .venv-jax/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt pytest
```

Core dependencies are NumPy, SciPy, Matplotlib, JAX, JAXlib, Optimistix, and Equinox. Optional integrations (`rocketcea` for variable chamber properties, `cadquery` for true revolved B-rep STEP) are not installed by `requirements.txt`. Run commands from the repository root; the project does not yet ship as an installable package.

8 Command-line usage

CLI pressure units are **bar** for P_c , **kPa** for P_a , and **millimeters** for R_t and manufacturing dimensions; internal APIs use SI units.

Preliminary Rao/TOP design.

```
.venv-jax/bin/python main.py \
  --propellant LOX/RP-1 \
  --Pc 45 --Pa 101.325 \
  --Rt 20 --epsilon 10 --length-pct 80 \
  --method bezier --no-plot \
  --output nozzle_profile.csv
```

Output is written to a versioned directory `builds/vNNN_YYYYMMDD_HHMMSS/`.

Size the throat from thrust.

```
.venv-jax/bin/python main.py \
  --propellant LOX/LCH4 \
  --Pc 60 --target-thrust 10000 --epsilon 12 --no-plot
```

The sizing relation is $R_t = \sqrt{F_{\text{target}} / (\pi C_{F,\text{actual}} p_c)}$.

Experimental Rao variational / MOC solve.

```
.venv-jax/bin/python main.py \
  --propellant LOX/RP-1 \
  --Pc 45 --Rt 20 --epsilon 10 \
  --method rao_variational_moc \
```

```
--rao-moc-n-control 12 --rao-moc-n-kernel 12 \
--rao-moc-max-nfev 200 --no-plot
```

This path can be expensive and remains experimental (`--rao-moc-max-nfev` defaults to 25). `--rao-moc-skip-moc` runs a faster residual-only study but prevents MOC reliability promotion.

Sweep and benchmark.

```
.venv-jax/bin/python main.py --propellant LOX/LCH4 \
--Pc 60 --Rt 25 --epsilon 10 --sweep epsilon 4 50 20 --no-plot

.venv-jax/bin/python main.py \
--benchmark-case lea_top_schomberg_2014 \
--benchmark-method bezier \
--benchmark-report builds/benchmarks --no-plot
```

Sweep variables are `epsilon`, `Pc`, `Rt`. Benchmark manifests are `lea_top_schomberg_2014`, `rao_scarfed_moc_1990`, and `vulcain_s1_separation_similarity`; each metric is classified `strict`, `xfail`, or `report`.

9 Python API

Baseline contour and performance.

```
from raosim.engine import compute_engine_performance
from raosim.nozzle_geometry import bell_nozzle_contour
from raosim.propellants import get_propellant

prop = get_propellant("LOX/RP-1")
contour = bell_nozzle_contour(
    Rt=0.020, epsilon=10.0, length_pct=80.0,
    method="bezier", gamma=prop.gamma,
)
performance = compute_engine_performance(
    Pc=4.5e6, Pa=101325.0, Rt=0.020, epsilon=10.0, prop=prop,
)
```

Design-gated workflow.

```
from raosim.design import DesignInput, ThermoSpec, design_nozzle_v2

result = design_nozzle_v2(DesignInput(
    thermo=ThermoSpec(mode="constant_gamma", propellant_name="LOX/RP-1"),
    Pc=8.0e6, Rt=0.020, epsilon=8.0,
    method="bezier", mode="preliminary",
))
print(result.performance.thrust)
print(result.gate_report.to_dict())
```

The mode `validated` uses the stricter schema and gate set; it does **not** set `hardware_qualified = True`.

Rao BVP and sensitivities.

```
from raosim.rao_variational import RaoSolverConfig, solve_rao_bvp
from raosim.jax import rao_sensitivities
```

```

config = RaoSolverConfig(
    Rt=0.020, epsilon=10.0, gamma=1.4, length_pct=80.0,
    solver_backend="jax", formulation="characteristic",
)
solution = solve_rao_bvp(config)
sens = rao_sensitivities(config, solution=solution)
print(solution.reliability, solution.residuals.max_scaled)
print(sens.cf, sens.condition_number)

```

Pass `solver_backend="numpy"` to use the legacy SciPy path.

10 Outputs

Normal CLI runs create `builds/vNNN_YYYYMMDD_HHMMSS/` and may contain: a contour CSV in meters; a binary STL inner surface or closed solid; a STEP solid (CadQuery revolved B-rep when available, else a faceted AP214 fallback); an Inventor conversion manifest pointing to the authoritative STEP file; design-gate JSON and v2 physics-screening sections; sweep CSV or benchmark JSON/Markdown reports; and a human-readable `metadata.txt`. Native Autodesk Inventor IPT writing is not implemented; bolt patterns, injector interfaces, throat inserts, tolerances, and weld/braze allowances are metadata placeholders, not modeled solid features.

11 Validation and reference evidence

The repository deliberately separates *software verification* from *physical validation*.

11.1 Software and mathematical verification

The test suite covers gas-dynamics identities and inverse relations; Bézier, conical, chamber, curvature, export, and design-workflow behavior; axisymmetric characteristic pairing and source terms; transonic Kliegel–Levine coefficients and NASA-visible-source semantics; characteristic topology, mass conservation, BDE closure, wall tangency, and crossing checks; NumPy/JAX primitive and residual parity; exact-Jacobian convergence, kernel-march parity, solved- θ_B derivatives, and C_F sensitivities; literature-manifest parsing with strict/report/xfail policies; NASA/JHU legacy-output and Tecplot parsing; and plotting/diagnostic metadata.

11.2 NASA/JHU reference port

`Three-Dimensional-Nozzle-Design-Code-master/` vendors the Rice/JHU 2D MOC, streamline-tracing, and 3D MOC source plus sample outputs. The active Python port implements the initial throat line and kernel march; arc-wall, interior, special-wall, and axis unit processes; mass flow along right-running characteristics; point- D selection, point- E integration, and fixed-/free-end closure; the θ_B secant solution; BDE-region and wall-contour construction; and explicit `RaoTopology` objects for TT' , B , BF , D , BD , DE , E , and the wall streamline.

The checked-in M3.5 perfect-nozzle wall comparison is within the repository's 10^{-3} RMS regression gate (currently $\approx 1.8 \times 10^{-4}$ against `wall.out`). However, the generator provenance of the historical `TT'.out` fixture is **unresolved**: the visible `MOC_GridCalc_BDE.cpp` source-port workflow is the canonical reference track, and the orphaned M3.5Perf fixture is a diagnostic overlay only — not promotion authority. Consequently the public solver does not label any contour `NASA_REFERENCE_MATCHED`.

11.3 Reliability labels

The research code defines an ordered `ContourReliability` vocabulary:

```
geometric_approximation → moc_compatible → rao_variational_residual_solved →
    nasa_reference_matched → benchmark_validated → cfd_checked →
        experimentally_validated.
```

This is a *vocabulary*, not a claim that every level has been reached. The `rao_variational_moc` solver assigns its level from per-solve gates: a clean solve passing the BVP residual, MOC closure, valid-region, and thrust-sanity gates reaches `rao_variational_residual_solved`. Promotion to `benchmark_validated` is gated behind the module flag `BENCHMARK_VALIDATED_AT_RELEASE`, which is **False** in the current code, so no public contour is benchmark-validated, NASA-reference-matched, CFD-checked, or experimentally validated by this repository.

12 Repository layout

```
main.py Primary interactive/batch CLI
raosim/ Active Python package
    gas_dynamics.py Isentropic, area-Mach, Prandtl-Meyer relations
    engine.py Ideal performance (Cf, Isp, c*, m_dot)
    propellants.py Combustion-product property database
    nozzle_geometry.py Rao/TOP Bezier geometry + chart tables
    conical.py Conical reference contour + divergence factor
    chamber_geometry.py L*/CR chamber + convergent sizing
    design.py High-level schemas, gates, artifacts
    validation.py Design-status labels and engineering gates
    physics.py BL, thermal, cooling, structural screens
    separation.py Summerfield / Kalt-Badal / Schmucker screens
    wall_pressure.py Quasi-1D wall-pressure distribution
    atmosphere.py / altitude_performance.py / trajectory.py
    moc.py General axisymmetric MOC march + starting lines
    transonic_kernel.py Kliegel-Levine transonic series
    nasa_moc.py NASA/JHU-style kernel and BDE topology port
    moc_topology.py RaoTopology objects (TT', B, BD, D, DE, E, wall)
    rao_optimizer.py Direct MOC wall optimization
    rao_variational.py Rao functionals, validity, and global BVP
    rao_residuals.py Characteristic residual primitives
    jax/ Differentiable primitives, residuals, kernel
        march, LM solve, and sensitivities
    cea.py Optional RocketCEA thermochemistry
    benchmarks.py Literature benchmark runner
    benchmark_data/ Manifests and digitized reference curves
    plotting.py Geometry, field, topology, residual plots
    export.py CSV, STL, STEP, IPT-manifest export
tests/ Unit, regression, parity, benchmark tests
scripts/ Research diagnostics and artifact generators
docs/ NASA provenance, topology, and audit notes
latex-report/ Mathematical reference report and figures
propulsion_texts/ Local source literature used by the project
Three-Dimensional-Nozzle-Design-Code-master/
    Vendored Rice/JHU reference source and outputs
```

13 Known limitations

1. **No hardware qualification.** All results require independent analysis and test evidence.
2. **Constant-property nozzle flow.** CEA does not yet drive variable γ , composition, or transport through the MOC field.
3. **Inviscid MOC.** Boundary-layer growth is a separate screen, not coupled into marching or optimization.
4. **Incomplete separation physics.** Empirical onset criteria only.
5. **Experimental exact-Rao path.** Mathematically auditable and tested, but release-level literature promotion is disabled (`BENCHMARK_VALIDATED_AT_RELEASE = False`).
6. **Chart vs. exact-variational angles.** Rao 1960 chart angles are parabola-fit data; exact deltas are recorded, not forced to zero.
7. **Partial differentiability.** The complete start-line-to-final-wall design map and total design derivatives are unfinished.
8. **Low-order thermal/structural models.** Screens, not CHT, coolant-network, fatigue, creep, or FEA.
9. **Simplified CAD.** Channels, bolt holes, injector faces, inserts, and welds are not solid-modeled.
10. **No package metadata or stable API guarantee.**

14 Remaining work

1. Complete and publish the exact-variational chart/reference sweep, define accepted exact-vs-TOP deltas, and set criteria for flipping `BENCHMARK_VALIDATED_AT_RELEASE = True`.
2. Make the live differentiable θ_B solve routine and finish total implicit design derivatives w.r.t. R_t , ε , $L\%$, γ , and ambient pressure.
3. Differentiate or replace the remaining start-line and BDE/final-wall steps so sensitivities reach manufacturable wall coordinates.
4. Finish source-port provenance and public reliability wiring for the NASA/JHU reference workflow, including the historical TT' fixture.
5. Add variable-property frozen/equilibrium thermochemistry and transport to the nozzle flowfield.
6. Couple viscous displacement, wall pressure, separation, and side-load physics to the contour; validate against experiments and CFD.
7. Add independent axisymmetric Euler/RANS CFD comparisons with quantitative gates for C_F , wall pressure, exit Mach, and flow angle.
8. Replace thermal/cooling/structural screens with conjugate heat transfer, coolant models, temperature-dependent materials, and structural analysis.
9. Promote CAD from a revolved screening solid to a manufacturing definition with channels, interfaces, fasteners, and tolerances.
10. Retire or isolate legacy paths, add package metadata, and publish a stable API and reproducible release artifacts.

15 References

Primary literature in `propulsion_texts/` and reference notes under `docs/` and `raosim/benchmark_data/`:

- G. V. R. Rao, *Exhaust Nozzle Contour for Optimum Thrust*, Jet Propulsion, 1958.
- G. V. R. Rao, *Approximation of Optimum Thrust Nozzle Contour*, ARS J. 30(6), 1960 — the TOP θ_n/θ_e design charts.
- G. V. R. Rao, *Recent Developments in Rocket Nozzle Configurations*, ARS J. 31(11), 1961 — γ -insensitivity of the optimal contour.
- Rao, Beck & Booth, *Rao Variational Optimum Bell Nozzle: A Design Compendium*, AIAA 99-2584, 1999 — stationarity Eqs. (1)–(2), validity boundary.
- NASA SP-8120, *Liquid Rocket Engine Nozzles* — chart reproduction, separation and wall-pressure guidance.
- Rice, *2D and 3D Method-of-Characteristics Tools for Complex Nozzle Development*, JHU/APL RTDC-TPS-481, 2003.
- Kliegel & Levine, *Transonic Flow in Small Throat Radius of Curvature Nozzles*, 1969.
- J. Östlund, *Flow Processes in Rocket Engine Nozzles*, KTH, 2002 — Rao valid-region discussion.
- R. Stark, *Flow Separation in Rocket Nozzles — An Overview*, 2009.
- J. D. Anderson, *Modern Compressible Flow* — isentropic relations, Prandtl–Meyer, MOC compatibility (§11.4).
- M. J. Zucrow & J. D. Hoffman, *Gas Dynamics*, Vol. 2, Ch. 17 — axisymmetric characteristics.
- G. P. Sutton & O. Biblarz, *Rocket Propulsion Elements* — performance relations.

Where any status statement in historical research notes conflicts with the executable code or the current test suite, **the code and tests are authoritative**. See `latex-report/raosim_reference.pdf` for the extended mathematical reference and `JAX_DIFFERENTIABLE_PLAN.md` for the differentiable-solver development record.